

Balancing Climate Ambition and Price Stability: The Case of South Africa's Carbon Tax

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Abstract

This paper examines whether South Africa's carbon tax, introduced in 2019 as part of its broader decarbonization strategy, has affected consumer prices. Using the Generalized Synthetic Control Method and quarterly data for 2015–2022, we estimate the causal impact of the tax on headline and disaggregated consumer price indices. The results show no statistically significant effect on overall inflation, while a small cumulative increase of about one percent is observed in energy prices relative to a counterfactual without the tax. These findings suggest that South Africa's carbon tax has altered relative prices within the economy without generating aggregate inflationary pressures. The results highlight how careful policy design—featuring phased implementation, generous transitional allowances, and integration with existing levies—can mitigate short-term cost pass-through while maintaining credible long-term price signals. The South African experience provides valuable lessons for emerging economies pursuing carbon pricing under inflation and growth constraints.

Keywords: Carbon tax; Greenflation; Energy policy; South Africa; Emerging economies; Synthetic control method.

JEL classification codes: E31, H23, Q43, Q54, Q58

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1 Introduction

Carbon pricing has emerged as a central policy instrument for mitigating greenhouse gas (GHG) emissions and promoting the transition to low-carbon growth. As of 2023, over 70 carbon pricing initiatives—either carbon taxes or emissions trading schemes (ETS)—are in operation worldwide, covering roughly one-quarter of global GHG emissions (World Bank, 2023a). While carbon pricing has proven effective in reducing emissions and stimulating cleaner production, it has also raised concerns about potential inflationary pressures—often referred to as “greenflation” (IMF, 2022; Schnabel, 2022; Konradt and Weder di Mauro, 2023; Konradt et al., 2024; Moessner, 2025). These concerns have gained prominence in the wake of rising global energy prices, prompting questions about whether carbon pricing could exacerbate cost-of-living increases, particularly in developing and economies where energy constitutes a large share of household expenditure and production costs (García and Suárez-Varela, 2022).

In theory, the impact of carbon pricing on consumer prices is ambiguous. On the supply side, carbon taxes can increase production costs for energy-intensive sectors, potentially leading to higher prices if these costs are passed through to consumers (Andersson et al., 2020; ICC, 2023). On the demand side, higher carbon prices can reduce consumption and investment, dampening inflationary pressures (ICC, 2023). The overall effect depends on country-specific factors such as tax design, the level of exemptions or free allowances, energy market structures, and exchange rate dynamics.

This paper contributes to the ongoing debate on the macroeconomic consequences of carbon pricing by providing the first causal, ex-post evidence on the inflationary effects of a carbon tax in an emerging-economy context. South Africa offers a particularly relevant case. As the largest emitter of greenhouse gases (GHGs) in Africa, the country introduced a carbon tax in June 2019 as part of its commitment to achieving net-zero emissions by 2050. The tax was designed with a phased implementation and transitional measures including a relatively low initial statutory rate of R120 per tonne of CO₂e, extensive tax-free allowances of up to 95% depending on the sector, and the

deductibility of pre-existing fuel and electricity levies from carbon tax liabilities (National Treasury, Republic of South Africa, 2024, 2023). The tax also exempts certain sectors, such as agriculture and waste, and temporarily excludes Eskom, the national electricity utility, from liability during the first phase. Together, these design features make South Africa's carbon tax a useful natural experiment for assessing whether well-calibrated carbon pricing can advance decarbonization goals without generating inflationary side effects.

Using quarterly data from 2015 to 2022, we apply the Generalized Synthetic Control Method (GSCM) (Xu, 2017) to estimate the causal impact of the carbon tax on headline, core, energy, and food consumer price indices (CPIs). This approach constructs a synthetic counterfactual for South Africa's consumer prices in the absence of the tax while controlling for unobserved heterogeneity and global shocks, such as the COVID-19 pandemic and the Russia–Ukraine conflict. Robustness checks using the Synthetic Difference-in-Differences estimator (Arkhangelsky et al., 2021) and a Sub-Saharan Africa-only donor pool confirm the stability of our results.

The findings indicate that South Africa's carbon tax has not produced statistically significant increases in headline inflation relative to the counterfactual. The effects are limited to a small cumulative rise—about 1%—in energy prices, consistent with partial cost pass-through in carbon-intensive sectors. This muted impact reflects the policy design of the first phase, which effectively cushioned consumers from price shocks while embedding a carbon price signal in the economy.

From a policy perspective, these results suggest that gradual implementation, generous allowances, and integration with existing fiscal instruments can mitigate short-term inflationary risks while preserving long-term environmental incentives. As South Africa enters the second phase of its carbon tax—marked by higher effective rates and reduced allowances—maintaining transparency, institutional coordination, and targeted support will be essential. More broadly, South Africa's experience provides evidence that well-designed carbon pricing can advance climate goals without compromising price stability in emerging economies.

This paper contributes to three interrelated strands of the literature. First, it extends empirical research on tax incidence, cost pass-through, and the price effects of carbon taxation by providing a causal, ex-post analysis of a carbon tax in an emerging-economy context. Prior studies have shown substantial pass-through of energy and emissions costs in industrial settings and near-complete pass-through to electricity prices (e.g., [Fabra and Reguant, 2014](#); [Ahn, 2025](#)), yet evidence on whether carbon pricing generates broad-based consumer inflation remains limited and mixed. Most existing analyses focus on advanced economies and emissions trading systems rather than carbon taxes ([Berthold et al., 2023](#); [Konradt and Weder di Mauro, 2023](#); [Konradt et al., 2024](#); [Moessner, 2025](#)).

Second, this study contributes to the literature on carbon-tax design and its macroeconomic implications in emerging markets. While most prior work examines developed-country experiences, this paper complements and contrasts with earlier ex-ante simulations of South Africa’s carbon tax that assessed its macroeconomic ([Alton et al., 2012](#); [Nong, 2020](#)) and distributional ([Okonkwo, 2021](#); [Dorband et al., 2019](#)) impacts. By applying the Generalized Synthetic Control Method (GSCM) to South Africa, we provide rigorous, quasi-experimental evidence on how a comprehensive carbon tax affects headline, core, food, and energy consumer prices.

Third, the paper advances methodological and policy debates on greenflation and the macroeconomic risks of climate policy. Through the use of the GSCM, it contributes to the empirical toolkit for evaluating environmental policies in data-limited settings and demonstrates that the inflationary impact of South Africa’s carbon tax is largely confined to direct energy-related channels, with no evidence of broader spillovers. This finding supports the view that carbon pricing primarily shifts relative prices rather than overall price levels ([Konradt and Weder di Mauro, 2023](#); [Konradt et al., 2024](#); [Moessner, 2025](#); [Andersson, 2019](#)).

The remainder of the paper is organized as follows. Section 2 provides background on South Africa’s carbon tax design and implementation. Section 3 describes the empirical strategy and data. Section 4 presents the results, while Section 5 reports

robustness checks. Section 6 concludes with policy implications.

2 Background on South Africa's Carbon Tax

South Africa is the largest emitter of GHGs on the African continent, with emissions primarily driven by a coal-dominated energy sector (Figure 1). In response to the global and domestic imperatives of decarbonization, the government enacted the Carbon Tax Act No. 15 of 2019, marking South Africa as the first African country to implement an explicit price on carbon. The carbon tax forms a cornerstone of the country's broader climate strategy, aimed at achieving its nationally determined contribution (NDC) targets of reducing GHG emissions to between 398–510 MtCO₂e by 2025, 350–420 MtCO₂e by 2030, and reaching net-zero emissions by 2050.¹ The tax operates under the *polluter-pays* principle, designed to internalize the social cost of carbon and incentivize firms and consumers to shift toward cleaner energy sources.

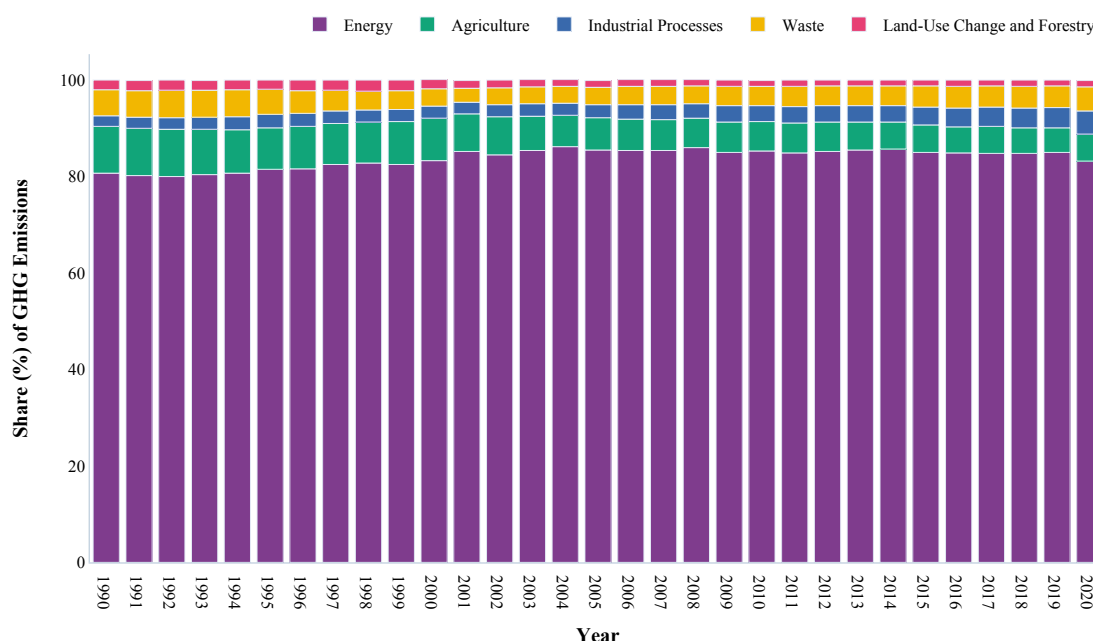


Figure 1: Sectoral Composition of South Africa's Historical GHG Emissions

Notes: Author's construct based on World Resources Institute (2022)

¹<https://unfccc.int/sites/default/files/NDC/2022-06/South%20Africa%20updated%20first%20NDC%20September%202021.pdf>

2.1 Policy Design and Phased Implementation

The carbon tax is being implemented in two phases. The first phase runs from June 1, 2019, to December 31, 2025,² while the second phase will span from January 1, 2026, to December 31, 2030. The tax currently applies to key sectors responsible for GHG emissions, including fuel combustion, industrial processes, and fugitive emissions, with agriculture, forestry, land use, and waste sectors temporarily excluded. The initial tax rate was set at R120 (approximately USD 6.30) per tonne of CO₂e, with an annual escalation of consumer price inflation plus two percent until 2022, and thereafter in line with inflation alone. As of January 2024, the rate stands at R190 (approximately USD 10) per tonne.³

A central feature of the tax design is the inclusion of generous tax-free allowances during the first phase, ranging from 60% to 95% depending on sectoral characteristics (see Figure 2). These allowances include a basic tax-free threshold (60–75%), additional relief for trade-exposed sectors (10%), performance-based incentives (up to 5%), carbon-offset allowances (up to 10%), and a 5% carbon-budget allowance for firms participating in mandatory emissions reporting. These provisions effectively reduce the initial tax burden while providing firms with a predictable pathway to adjust operations and invest in low-carbon technologies. Eskom, the country's state-owned electricity producer and largest single emitter, was temporarily exempted from paying the tax during the first phase—further limiting immediate pass-through effects to electricity prices. Due to the tax-free allowances, the effective rate could be as low as ZAR 6 (approximately US\$0.40) per ton of CO₂ emissions (Steenkamp, 2022).

²Originally planned to end in December 2022, the first phase was extended during the 2022 Budget Review.

³<https://www.treasury.gov.za/documents/National%20Budget/2024/review/Chapter%204.pdf>

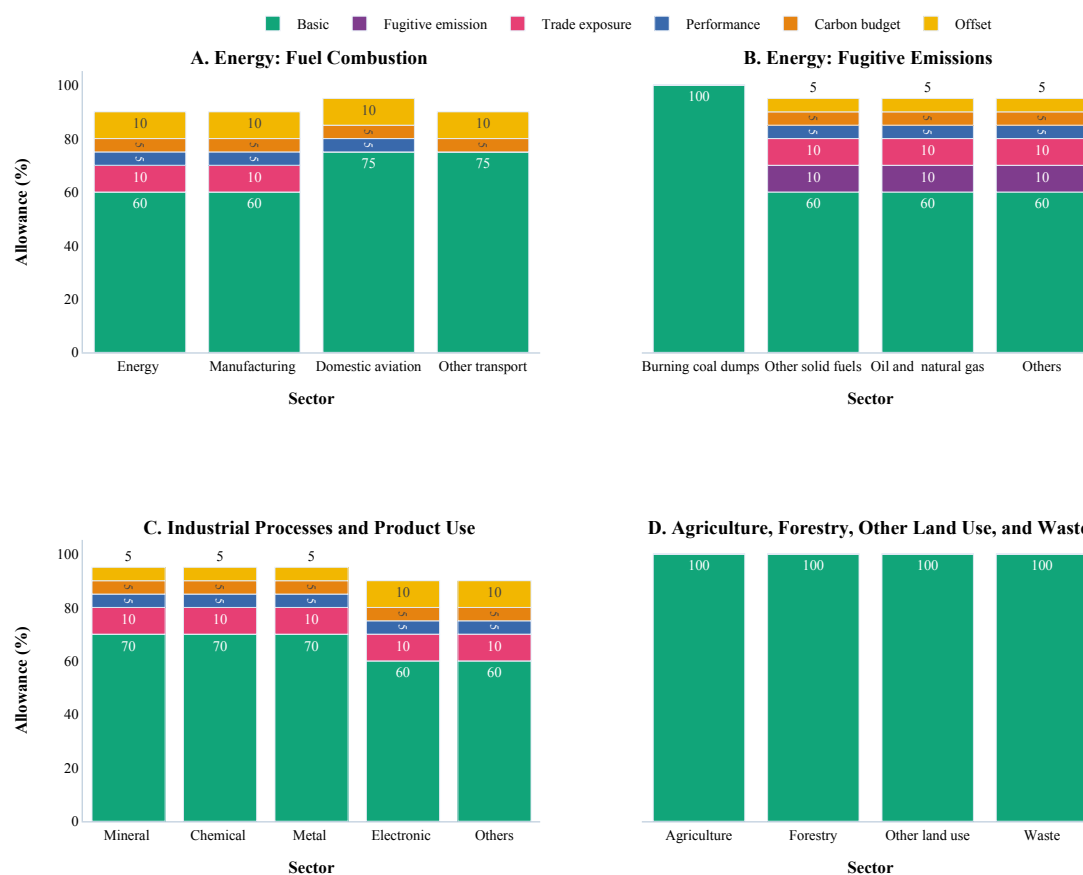


Figure 2: Sectoral Composition of South Africa's Historical GHG Emissions

Source: Author's Construct based on [Government of the Republic of South Africa \(2019\)](#)

Although the initial carbon tax rate is below the estimated global benchmark of USD 40–80 per tonne needed to meet the Paris Agreement goals, the government has announced plans to progressively strengthen the price signal by reducing allowances after 2026 and scaling up rates toward international standards ([National Treasury, Republic of South Africa, 2024](#)). This phased approach reflects a strategy of gradualism—balancing the urgency of climate action with the need to safeguard economic stability and competitiveness.

2.2 Integration with Other Environmental Levies

The carbon tax is part of a broader environmental fiscal framework that includes existing energy-related levies. To ensure policy coherence and avoid overlapping charges,

firms are allowed to deduct certain pre-existing levies—specifically, the petroleum and diesel levies, and the electricity levy—from their carbon tax liabilities. This provision prevents double taxation of emissions sources already covered by environmental charges under the Customs and Excise Act and aligns the carbon tax with complementary fiscal instruments ([National Treasury, Republic of South Africa, 2023](#)).

The petroleum and diesel levies, introduced in 2003, fund road infrastructure and public transport while encouraging fuel efficiency. The electricity levy, implemented in 2009, applies to electricity generated from non-renewable sources—primarily coal—and aims to promote energy diversification. Together, these levies and the carbon tax form an integrated environmental fiscal system that internalizes the social cost of energy consumption while maintaining competitiveness through coordinated deductions. This design mitigates short-term adjustment costs and reinforces the long-term transition toward cleaner energy sources.

2.3 Administrative Design and Compliance Mechanisms

The carbon tax is administered by the South African Revenue Service (SARS) under the Customs and Excise Act, leveraging existing tax collection systems to minimize compliance costs and enhance administrative efficiency ([National Treasury, Republic of South Africa, 2023](#)). Taxable entities must calculate and report their emissions annually in accordance with the Department of Forestry, Fisheries and the Environment's (DFFE) greenhouse gas reporting regulations, ensuring consistency between fiscal and environmental data systems. This institutional coordination enhances the credibility of emissions reporting and aligns fiscal enforcement with South Africa's national inventory commitments.

The phased implementation strategy also provides compliance flexibility during the initial years. Firms can use verified carbon offsets—up to 10% of their liability—to reduce their effective tax payments. Eligible offset projects must meet stringent verification standards, including those under the Clean Development Mechanism (CDM) and the Verified Carbon Standard (VCS), ensuring that emissions reductions are real,

additional, and verifiable. This design enhances environmental integrity while providing cost-effective abatement options for emitters. The system of allowances, offsets, and phased escalation thus reflects a pragmatic balance between policy ambition, administrative capacity, and economic feasibility.

2.4 Revenue Use and Fiscal Linkages

Carbon tax revenues are collected by SARS and deposited into the National Revenue Fund in accordance with the Public Finance Management Act. While revenues are not formally earmarked, the National Treasury has adopted a de facto recycling approach, channeling funds toward programs that support the energy transition and mitigate adverse socioeconomic effects (National Treasury, Republic of South Africa, 2024, 2023). These include energy-efficiency tax incentives under Section 12L of the Income Tax Act, public transport and renewable energy investments, and the Energy Transition and Climate Change Response Program.

Revenue recycling serves multiple objectives: cushioning low-income households from energy price increases, preserving industrial competitiveness, and supporting green investment. This approach aligns with international best practice, which recommends using carbon revenues to enhance both the political acceptability and economic efficiency of carbon pricing (e.g., Marron and Toder, 2014; Metcalf, 2021; Parry et al., 2021). As South Africa moves into the second phase of implementation, the National Treasury has indicated plans to improve transparency in revenue allocation and strengthen the fiscal linkages between carbon tax receipts and climate-related expenditure (National Treasury, Republic of South Africa, 2024). This evolution will help integrate the carbon tax more fully into the country's broader strategy for sustainable public finance and climate resilience.

2.5 Summary: Policy Design and Analytical Motivation

Overall, South Africa's carbon tax reflects a deliberate and integrated approach to environmental fiscal reform. Its coherence with pre-existing levies, alignment of ad-

ministrative and environmental reporting systems, and pragmatic revenue recycling framework collectively demonstrate a policy design that prioritizes both feasibility and credibility. The generous transitional provisions and offset mechanisms reduce immediate inflationary and competitiveness risks while maintaining a credible long-term price signal for decarbonization.

By embedding carbon pricing within South Africa’s institutional and fiscal architecture, the government has strengthened the foundation for a durable and adaptive climate policy. This coherent policy structure also provides an empirical opportunity to test whether such a carefully designed carbon tax—characterized by gradual implementation, generous allowances, and offset flexibility—has nonetheless exerted measurable effects on consumer prices. The next section turns to this question using a quasi-experimental approach based on the Generalized Synthetic Control Method.

3 Methodology and data

3.1 Empirical Strategy

We estimate the causal impact of South Africa’s carbon tax on consumer prices using the Generalized Synthetic Control Method (GSCM) developed by [Xu \(2017\)](#). The GSCM is particularly suitable for policy evaluations in settings where randomized experiments are infeasible and where treatment occurs for a single or few units—such as South Africa’s national carbon tax. It constructs a synthetic counterfactual from a weighted combination of untreated countries, allowing for both unit-specific and time-varying unobserved heterogeneity.

3.1.1 Model Framework

Following [Xu \(2017\)](#), the outcome for country j at time t is modeled as:

$$Y_{jt} = \alpha_0 + \delta_{jt}\text{Tax}_{jt} + \beta X'_{jt} + \lambda'_j f_t + \epsilon_{jt}, \quad (3.1)$$

where Y_{jt} denotes the logarithm of the consumer price index (CPI) for country j at time t ; Tax_{jt} is a treatment indicator equal to one for South Africa from 2019Q2 onward and zero otherwise; X_{jt} is a vector of observed covariates (real GDP growth and exchange rate); and ϵ_{jt} is an idiosyncratic error term. The term f_t represents unobserved common factors—such as global commodity prices or international shocks—while λ_j captures country-specific sensitivities to these factors.

The coefficient of interest, δ_{jt} , measures the treatment effect of South Africa’s carbon tax on consumer prices relative to the counterfactual. The GSCM estimates the counterfactual outcome, $\hat{Y}_{jt}^0$, by combining information from countries in the donor pool and modeling unobserved time-varying components through an interactive fixed effects (IFE) framework (Bai, 2009). This approach addresses a key limitation of the conventional synthetic control method, which assumes time-invariant unobservables, and provides standard errors and confidence intervals for statistical inference.

3.1.2 Identification and Inference

The identification strategy relies on the assumption that, conditional on the observed covariates and latent factors captured by the GSCM, trends in consumer prices in South Africa would have followed a similar path to that of the synthetic control in the absence of the carbon tax. The credibility of this assumption is assessed through the model’s ability to replicate South Africa’s pre-treatment CPI trajectory. A close pre-treatment fit indicates that the synthetic control provides a valid counterfactual.⁴

The estimated average treatment effect on the treated (ATT) at time t is expressed as:

$$\widehat{ATT}_t = Y_{jt}^1 - \hat{Y}_{jt}^0, \quad \text{for } t > T_0, \quad (3.2)$$

where T_0 denotes the final pre-treatment period. We report both time-varying treatment effects and cumulative effects over the post-treatment period, along with 95% confidence intervals derived from bootstrapped standard errors clustered at the

⁴Technical details on how the counterfactual is formulated can be found in [Appendix B](#)

country level.

3.1.3 Advantages of the GSCM

The GSCM offers several advantages for this analysis. First, it flexibly accounts for both country-specific and time-specific heterogeneity, which is critical given cross-country differences in monetary policy, fiscal responses, and exposure to global shocks such as COVID-19 or the Russia–Ukraine conflict. Second, unlike the traditional difference-in-differences (DiD) framework, the GSCM does not require the restrictive parallel-trends assumption. Finally, it provides formal statistical inference—something not available in the original synthetic control approach (Abadie and Gardeazabal, 2003; Abadie et al., 2010)—allowing for hypothesis testing and confidence interval construction.

Overall, this approach enables a transparent and data-driven evaluation of whether South Africa’s carbon tax led to measurable changes in consumer prices relative to a credible synthetic benchmark.

3.2 Data

3.2.1 Consumer Prices

Our main outcome variables are the headline consumer price index (CPI), core CPI, energy CPI, and food CPI. Quarterly data for these indices are sourced from the World Bank’s Global Inflation Database (Ha et al., 2023). The use of quarterly frequency offers two key advantages. First, it provides a sufficiently long pre-treatment period to construct a credible counterfactual trajectory of South Africa’s consumer prices prior to the introduction of the carbon tax. Second, quarterly data allow us to capture potential short-term or transitory price effects that might be obscured in annual data.

The donor pool for the synthetic control consists of developing countries with complete data for all four CPIs and without a carbon tax or emissions trading scheme (ETS) during the study period. This restriction ensures that the comparison countries remain unaffected by carbon pricing shocks. The final sample comprises 13 devel-

oping countries: Nigeria, Mauritius, Paraguay, Indonesia, the Philippines, Lao PDR, El Salvador, Malaysia, North Macedonia, Honduras, Guatemala, the Dominican Republic, and India. The sample period spans from 2015Q1 to 2022Q3, providing 17 pre-treatment quarters and 13 post-treatment quarters.

For robustness, we supplement the quarterly data with annual headline CPI data for Sub-Saharan African (SSA) countries obtained from [World Bank \(2024\)](#). This enables a separate analysis focusing exclusively on SSA economies where quarterly CPI data are unavailable.

3.2.2 Control Variables

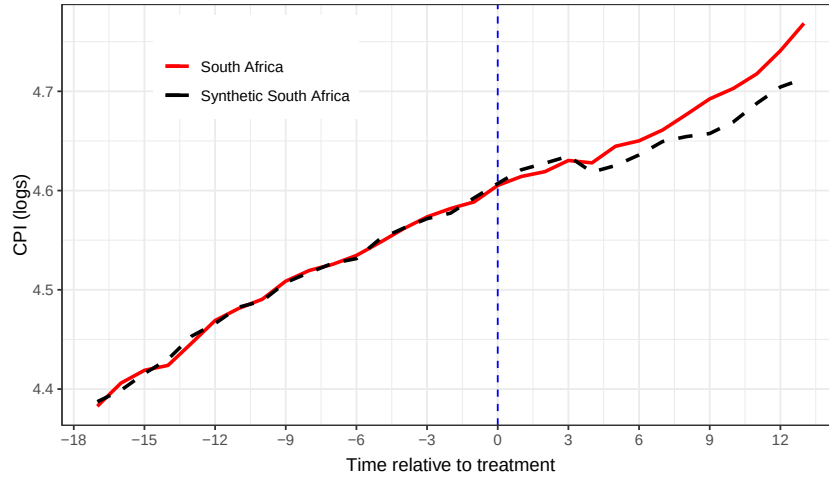
To account for other macroeconomic determinants of inflation, we include quarterly data on real GDP growth and the nominal exchange rate (local currency per US dollar) from the IMF's International Financial Statistics ([IMF, 2024](#)). In the robustness analysis based on annual data, we additionally control for trade balance and broad money supply, both sourced from the World Bank ([World Bank, 2024](#)). These controls capture demand-side and monetary factors that could influence consumer price dynamics independently of the carbon tax.

Variable definitions, sources, and units are provided in [Table A.1](#), while descriptive statistics are presented in [Table A.2](#).

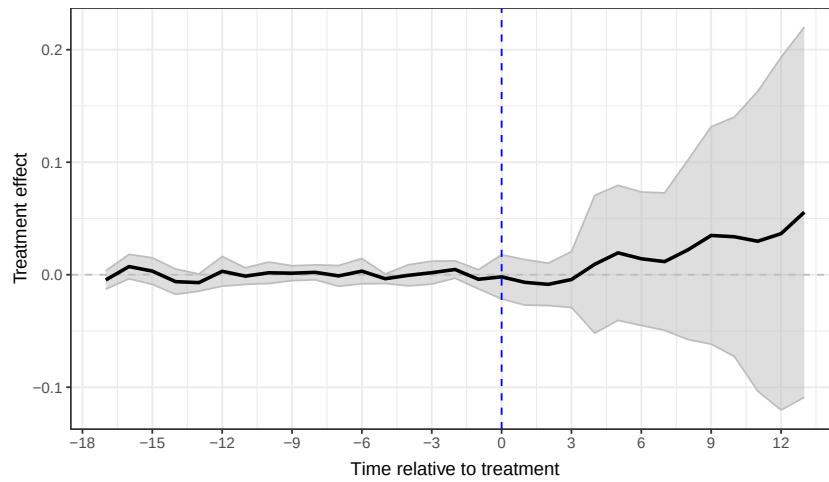
4 Results and Discussion

4.1 Impact on Headline Consumer Prices

[Figure 3](#) presents the estimated effects of South Africa's carbon tax on headline consumer prices using the GSCM. Panel (a) plots the observed trajectory of South Africa's headline CPI (solid red line) against the synthetic control (dashed black line), while the vertical blue line marks the introduction of the carbon tax in June 2019. The synthetic control reproduces South Africa's pre-treatment price trajectory closely, suggesting a well-constructed counterfactual.



(a) Treated and synthetic control



(b) Time Varying Treatment Effect

Figure 3: Impact of South Africa's Carbon Tax on Headline Consumer Prices

Notes: The figure presents the estimated effects of South Africa's carbon tax on the headline consumer price index (CPI) using the Generalized Synthetic Control Method (GSCM) proposed by Xu (2017). Panel (a) depicts the trajectories of South Africa's headline CPI and its synthetic control, while Panel (b) shows the corresponding time-varying treatment effects. The shaded gray area represents the 95% confidence interval around the estimated effects.

Following the tax implementation, South Africa's headline CPI remains almost identical to its synthetic counterpart for roughly three quarters, after which a slight divergence appears. Although the observed CPI begins to exceed the counterfactual, this difference remains statistically insignificant, as shown in Panel (b), which plots the time-varying treatment effect and its 95% confidence interval. In percentage terms, the deviation does not exceed 0.5%, and the confidence band consistently includes zero.

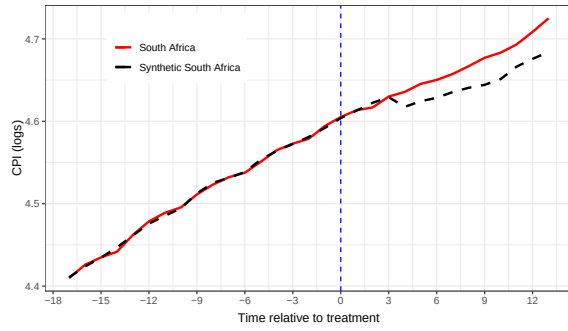
This finding implies that the introduction of the carbon tax did not have a statistically significant inflationary effect on aggregate consumer prices. The result aligns with recent evidence from the Euro area (Konradt and Weder di Mauro, 2023; Konradt et al., 2024), Canada (Konradt and Weder di Mauro, 2023), and Sweden (Andersson, 2019), where carbon pricing reforms were also found to have negligible or transitory effects on overall inflation.

From a policy standpoint, this evidence indicates that South Africa’s carbon tax—despite its broad sectoral coverage—has been absorbed without disrupting macroeconomic price stability. This resilience can be attributed to three key factors: (i) the modest initial tax rate relative to international benchmarks, (ii) the extensive use of transitional tax-free allowances, and (iii) the temporary exemption of Eskom, the state-owned electricity producer, which delayed direct pass-through to electricity tariffs.

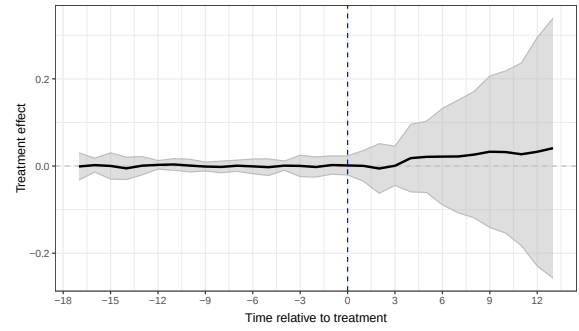
4.2 Disaggregated Impacts: Core, Energy, and Food Prices

To assess potential heterogeneity in price responses, Figure 4 presents the estimated effects of South Africa’s carbon tax on core, energy, and food consumer price indices (CPIs). Carbon pricing is expected to directly influence energy prices—through higher fossil fuel costs—and potentially affect food prices indirectly via increased production and transport costs if firms pass these on to consumers. Core CPI, which excludes energy, should remain largely unaffected if only direct effects operate (Baumeister et al., 2010; Gao et al., 2014).

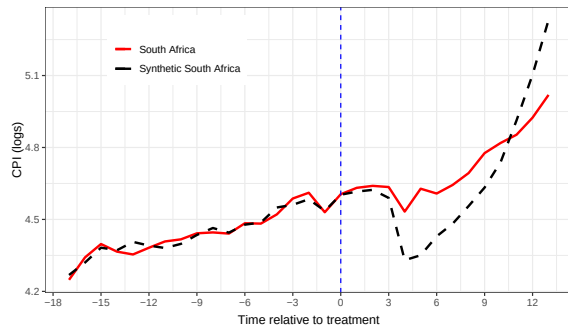
The results reveal a clear divergence in responsiveness across CPI components. For both the core and food CPIs, South Africa’s observed price trajectories closely track their synthetic controls before and after the carbon tax implementation. Estimated treatment effects are near zero and statistically insignificant throughout the post-treatment period. This pattern suggests the absence of indirect inflationary spillovers through input cost channels or food supply chains. In other words, the carbon tax did not propagate beyond directly affected energy-related sectors.



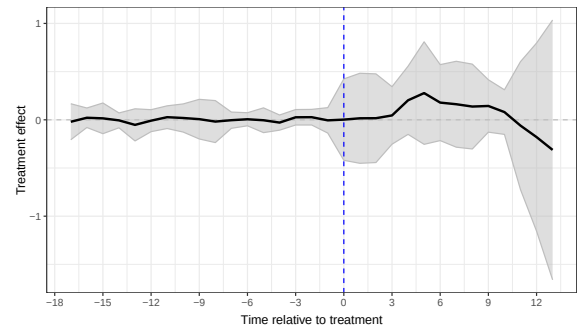
(a) Core CPI: Treated and Synthetic Control



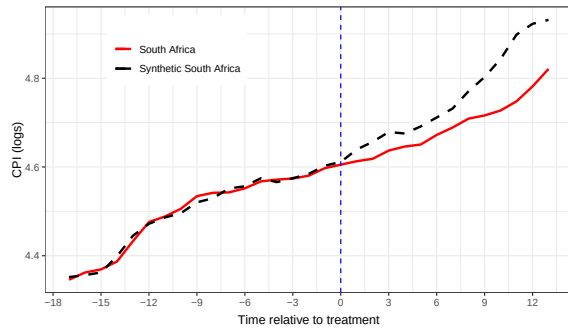
(b) Core CPI: Time Varying Treatment Effect



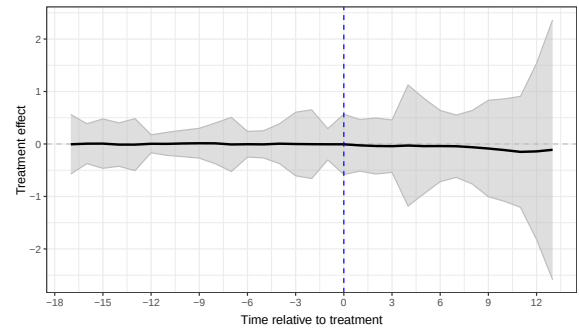
(c) Energy CPI: Treated and Synthetic Control



(d) Energy CPI: Time Varying Treatment Effect



(e) Food CPI: Treated and Synthetic Control



(f) Food CPI: Time Varying Treatment Effect

Figure 4: Impact of South Africa's Carbon Tax on Disaggregated Consumer Prices

Notes: The figure presents the estimated effects of South Africa's carbon tax on core, energy, and food consumer price indices, using the Generalized Synthetic Control Method proposed by Xu (2017). The shaded gray area represents the 95% confidence interval around the estimated treatment effects.

By contrast, the energy CPI exhibits a modest but more discernible adjustment. Around the third quarter following implementation, the observed energy CPI begins to rise slightly above the synthetic counterfactual, peaking at approximately a 1.2% cumulative increase by the eleventh quarter before tapering thereafter. Although the magnitude of this response is small, it is statistically distinguishable from zero in

several post-treatment quarters, indicating limited but measurable direct cost pass-through from the tax.

This component-level pattern aligns with theoretical expectations and international evidence: carbon pricing primarily affects directly taxed energy sectors, with negligible second-round effects on broader price indices (e.g. [Moessner, 2025](#)). Similar findings have been reported for the EU ETS, where energy prices exhibit the strongest response while core and food prices remain largely insulated ([Berthold et al., 2023](#); [Konradt et al., 2024](#)). The muted food and core responses in South Africa may reflect offsetting macroeconomic conditions—including exchange-rate movements, post-pandemic supply normalization, and competitive pricing pressures—that constrained firms' ability to pass on costs.

The modest and transient nature of the energy CPI increase also underscores the cushioning effect of South Africa's tax design. Generous first-phase allowances, offsets, and the phased escalation of rates limited cost transmission to consumers, while partial regulation of energy prices and subdued domestic demand further dampened potential inflationary effects. These dynamics are consistent with evidence that, in emerging markets, exchange-rate volatility, food supply shocks, and monetary policy play a more dominant role in shaping core inflation than environmental fiscal measures ([Ha et al., 2019a,b](#)).

Overall, the results reinforce the notion that carbon taxes primarily alter relative prices—raising the cost of carbon-intensive energy goods—without triggering generalized inflation. South Africa's experience illustrates that, under a well-calibrated and gradual framework, carbon pricing can reprice energy in line with environmental objectives while maintaining aggregate price stability.

4.3 Average and Cumulative Effects on Consumer Prices

To complement the time-varying estimates, we examine the average and cumulative impacts of South Africa's carbon tax on consumer prices over the post-treatment period. This allows us to assess whether any transitory effects accumulated into a

broader inflationary pattern following the tax introduction.

Table 1 reports the average treatment effects estimated using the GSCM, while Figure 5 illustrates the dynamic cumulative impacts over thirteen quarters after implementation. Consistent with earlier results, the estimates show no statistically significant average effect of the carbon tax on headline, core, or food CPI relative to the counterfactual. Point estimates are positive but small in magnitude, suggesting negligible aggregate inflationary consequences.

Table 1: GSCM Estimates of the Average Impact of South Africa's Carbon Tax on Consumer Prices

	Headline	Core	Energy	Food
carbon tax	0.017 (0.030)	0.019 (0.064)	0.050 (0.036)	-0.066 (0.239)
Country FE	✓	✓	✓	✓
Year-Quarter FE	✓	✓	✓	✓

Notes: The table reports the Generalized Synthetic Control Method estimates of the impact of South Africa's carbon tax on consumer prices. Robust standard errors, clustered at the country level, are in parentheses.

The cumulative analysis further confirms this pattern. As shown in Figure 5, cumulative deviations of headline, core, and food CPIs from their synthetic counterparts remain close to zero throughout the observation period. The confidence intervals widen gradually over time, reflecting reduced precision in long-run estimates, but the effects remain statistically indistinguishable from zero. These results indicate that the implementation of South Africa's carbon tax did not induce persistent upward pressure on consumer prices at the aggregate level.

The only notable and statistically significant cumulative impact emerges in the energy CPI, which begins to diverge from the counterfactual around the third quarter after the tax introduction. The cumulative difference peaks at roughly 1.2% in the eleventh quarter before stabilizing at about 1% after thirteen quarters. This modest magnitude implies partial cost pass-through limited to directly taxed energy goods, consistent with the tax's intended signal to reprice carbon-intensive consumption. Importantly, the absence of comparable effects in core or food CPI suggests that higher

energy costs did not spill over into broader consumer inflation.

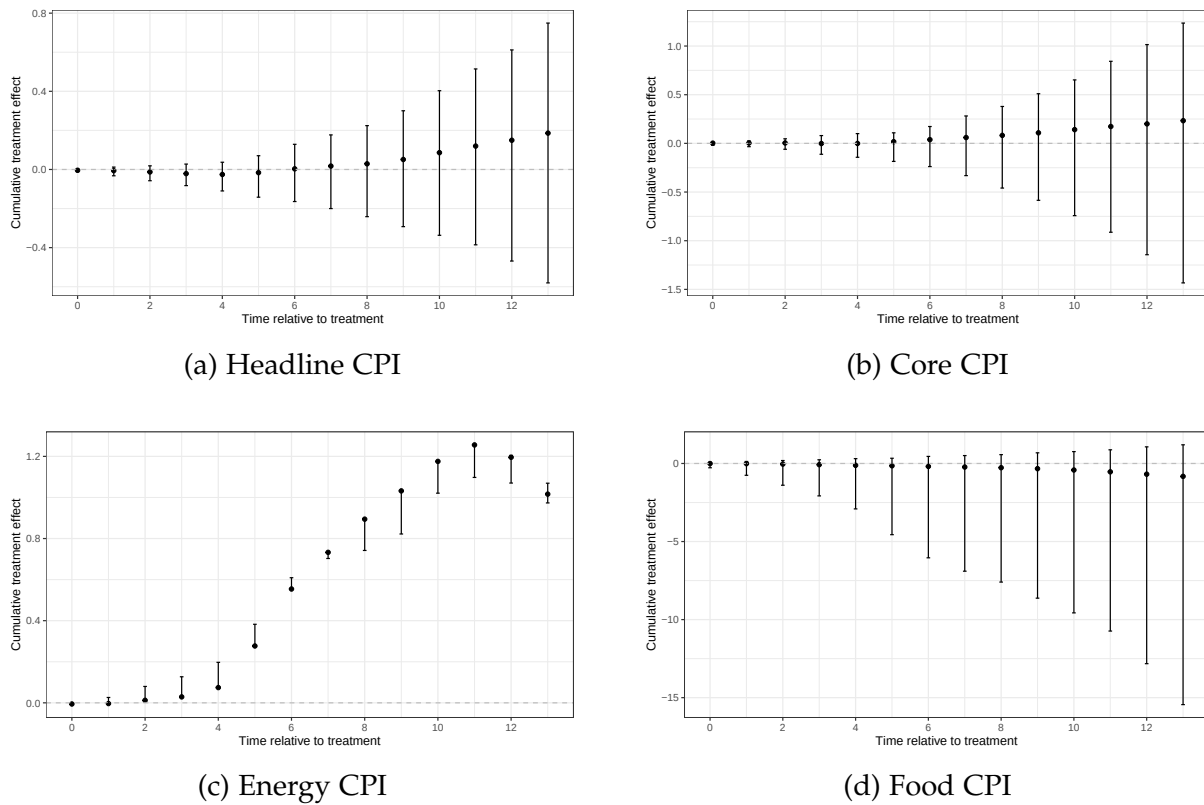


Figure 5: Cumulative Impact of South Africa's Carbon Tax on Consumer Prices

Notes: The figure presents the cumulative effects of South Africa's carbon tax on headline, core, energy, and food consumer price indices, estimated using the Generalized Synthetic Control Method proposed by [Xu \(2017\)](#). The horizontal bars represent the 95% confidence intervals around the estimated cumulative effects.

This finding aligns with evidence from advanced economies showing that carbon pricing primarily affects relative prices rather than aggregate inflation ([Konradt and Weder di Mauro, 2023](#); [Konradt et al., 2024](#); [Moessner, 2025](#)). It also resonates with theoretical models positing that carbon taxes influence prices chiefly through direct channels, with indirect effects constrained by substitution, market competition, and monetary policy responses ([Fullerton and Wolfram, 2012](#)). In South Africa's case, the design of the tax—characterized by extensive transitional allowances, the exclusion of electricity producer Eskom during the first phase, and a gradual escalation schedule—further dampened potential inflationary pressures.

Overall, these results reinforce the conclusion that South Africa's carbon tax has functioned as a relative price instrument rather than an inflationary shock. By al-

lowing sufficient adjustment time for firms and households, the policy managed to internalize environmental costs without compromising price stability. As the country enters the second phase of implementation, where allowances will tighten and the tax rate will rise, maintaining this balance will depend on continued transparency, complementary investments in energy efficiency, and targeted social protections for vulnerable households.

5 Robustness Checks

To assess the reliability of our baseline results, we perform two complementary robustness exercises. The first applies an alternative estimation framework—the Synthetic Difference-in-Differences (SDiD) method—while the second restricts the donor pool to SSA countries using annual data. Together, these tests evaluate whether our findings depend on model specification or sample composition.

5.1 Alternative Estimator: Synthetic Difference-in-Differences

Our baseline analysis employs the Generalized Synthetic Control Method (GSCM) (Xu, 2017), which flexibly models time-varying unobserved factors and permits classical inference. As a robustness check, we estimate the average treatment effects using the Synthetic Difference-in-Differences (SDiD) approach developed by Arkhangelsky et al. (2021). The SDiD framework combines the strengths of conventional difference-in-differences (DiD) and synthetic control methods by constructing weighted averages of control units and pre-treatment periods that best approximate the treated unit’s pre-intervention path. This design enhances the plausibility of the parallel-trends assumption and allows for robust inference under more relaxed data requirements.

The SDiD results, reported in Table A.3, confirm the baseline findings. Estimated coefficients for headline, core, and food CPIs remain statistically indistinguishable from zero, while the energy CPI shows a small positive but economically modest effect. The magnitudes are consistent with those obtained from the GSCM, indicating

that our main conclusions are not sensitive to the choice of estimation method. The alignment across methods reinforces the interpretation that South Africa’s carbon tax has not produced measurable inflationary effects beyond limited energy price adjustments.

5.2 Restricting the Donor Pool to SSA Countries

The second robustness check addresses potential concerns about the comparability of the donor pool. While the baseline model uses quarterly data from a diverse set of developing economies to ensure sufficient pre-treatment periods, this heterogeneity may introduce bias if price dynamics differ structurally from those of South Africa. To test this, we re-estimate the GSCM using annual data restricted to Sub-Saharan African countries with available macroeconomic indicators—namely Nigeria, Mauritius, Ghana, Burkina Faso, Cape Verde, Uganda, Zimbabwe, Kenya, and Madagascar—over the period 2000–2022.

Panel (a) of [Figure A.1](#) compares South Africa’s actual and synthetic headline CPI trajectories, while Panel (b) presents the corresponding treatment effects and Panel (c) displays cumulative impacts. Although the synthetic control replicates South Africa’s pre-treatment trend less precisely than in the quarterly specification, the estimated post-treatment effects remain statistically indistinguishable from zero. Both the point estimates and confidence intervals suggest no discernible inflationary response following the 2019 tax introduction. The cumulative impact of the carbon tax on headline CPI relative to the counterfactual also remains negligible, confirming the robustness of our baseline results to alternative donor pool construction.

Across both exercises, the empirical evidence consistently indicates that South Africa’s carbon tax did not trigger broad-based inflationary pressures. The small and transient response in energy CPI is robust to alternative estimation and sample restrictions, suggesting that any inflationary impacts are confined to directly taxed sectors and remain economically marginal. Taken together, these findings strengthen confidence in the causal interpretation of the baseline estimates and underscore the

stabilizing role of South Africa’s phased implementation, moderate tax rates, and transitional allowances. They further affirm that well-designed carbon pricing can coexist with macroeconomic stability even in emerging-market settings.

6 Conclusion and Policy Implications

6.1 Conclusion

This paper provides the first causal, ex-post evaluation of the impact of a carbon tax on consumer prices in an emerging country context. Using the Generalized Synthetic Control Method and quarterly data for 2015–2022, we assess the effects of South Africa’s 2019 carbon tax on headline, core, food, and energy consumer price indices. The results show that the introduction of the tax has not produced statistically significant increases in headline or core consumer prices relative to a counterfactual without the tax. The only discernible effect is a small cumulative rise of approximately 1% in the energy component of the CPI over three years, consistent with limited cost pass-through from carbon-intensive sectors.

These findings contribute to the broader debate on the macroeconomic implications of carbon pricing and the emerging notion of “greenflation.” Consistent with recent evidence for advanced economies (e.g., [Konradt and Weder di Mauro, 2023](#); [Konradt et al., 2024](#); [Moessner, 2025](#)), the results suggest that well-designed carbon taxes primarily alter relative prices rather than driving persistent aggregate inflation. In South Africa’s case, the muted price effects can be attributed to deliberate policy choices—namely, a moderate initial tax rate, extensive transitional allowances, and administrative integration with pre-existing fuel and electricity levies. Collectively, these features have allowed the government to embed a carbon price signal in the economy without compromising price stability.

6.2 Policy Implications

The results offer three main insights for policymakers in South Africa and other emerging economies.

First, *gradualism and predictability matter*. A phased introduction of carbon pricing with transitional flexibility allows firms and households to adjust to higher energy costs, mitigating inflationary spillovers and preserving political support.

Second, *design features determine macroeconomic outcomes*. Generous tax-free allowances, sectoral exemptions, and offset mechanisms can smooth distributional impacts and sustain competitiveness while maintaining incentives for long-term emissions reductions. In South Africa, these provisions—along with coordination between the National Treasury, the South African Revenue Service, and the Department of Forestry, Fisheries and the Environment—were crucial in preventing pass-through to consumer prices.

Third, *institutional coordination and transparency are essential*. Effective communication of tax schedules and revenue use, alongside integration with broader fiscal and energy policy, strengthens credibility and anchors inflation expectations. These lessons are particularly relevant as South Africa transitions into the second phase of its carbon tax (2026–2030), which will feature higher effective rates and declining allowances.

Our findings align with international policy guidance emphasizing that effective carbon pricing must balance environmental ambition with fiscal and social stability. Both the IMF and World Bank highlight that gradual implementation, predictable price trajectories, and integration with existing fiscal systems are essential to maintain macroeconomic stability and political support for carbon pricing (Parry et al., 2019, 2021; World Bank, 2022, 2023b). These institutions also emphasize the importance of revenue recycling—through energy-efficiency programs, social transfers, or tax reductions—to offset regressive effects and sustain public acceptance. South Africa’s carbon tax embodies many of these principles, demonstrating that, even in developing economies, well-calibrated policy design can align climate, fiscal, and distributional

objectives.

From a global perspective, South Africa’s experience demonstrates that carbon pricing can be both credible and non-inflationary in emerging markets when supported by sound design and gradual implementation. It aligns with international guidance from the IMF and World Bank, which emphasize policy coherence, revenue recycling, and social protection mechanisms as key to ensuring fiscal and political sustainability of carbon pricing frameworks. As countries operationalize carbon markets under Article 6 of the Paris Agreement, South Africa’s case illustrates that macroeconomic stability and climate ambition are not mutually exclusive. Well-calibrated carbon pricing can serve as a cornerstone of sustainable fiscal policy—advancing decarbonization goals while safeguarding growth and consumer welfare.

6.3 Limitations and Future Research

While this study provides causal evidence on the aggregate price effects of South Africa’s carbon tax, several limitations warrant further exploration. First, the analysis focuses on national-level consumer prices and does not capture distributional heterogeneity across income groups or regions. Future research could investigate how carbon taxation affects low-income households and energy-poor communities through both direct and indirect channels.

Second, firm-level data could be used to examine cost pass-through mechanisms, production adjustments, and potential competitiveness impacts. Third, as South Africa enters the second phase of the tax—with higher rates, declining allowances, and growing interactions with renewable energy policies—continuous empirical evaluation will be essential to monitor inflationary, fiscal, and distributional outcomes.

Finally, comparative studies across emerging and frontier economies—such as Colombia, Indonesia, and Chile—would deepen understanding of how institutional capacity, fiscal structure, and energy intensity shape the macroeconomic consequences of carbon pricing. Such work would help clarify how developing countries can design climate policies that jointly achieve emissions mitigation, fiscal resilience, and price

stability.

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Appendix A: Additional Tables and Figures

Table A.1: Variable Definitions and Data Sources

Variable	Frequency	Definition	Source
Headline CPI	Quarterly and Annual	Cost to the average consumer of acquiring a basket of goods and services index	Ha et al. (2023); World Bank (2024)
Core CPI	Quarterly	Cost to the average consumer of acquiring a basket of goods and services index	Ha et al. (2023)
Energy CPI	Quarterly	Cost to the average consumer of acquiring energy services	Ha et al. (2023)
Food CPI	Quarterly	Cost to the average consumer of acquiring a basket of food and non-alcoholic beverages	Ha et al. (2023)
Exchange Rate	Quarterly and Annual	National currency per U.S. dollar (period average)	IMF (2024)
GDP	Quarterly and Annual	Real gross domestic product, seasonally adjusted	IMF (2024)
Trade Balance	Annual	Net trade in goods and services	World Bank (2024)
Broad Money	Annual	Percent of GDP	World Bank (2024)

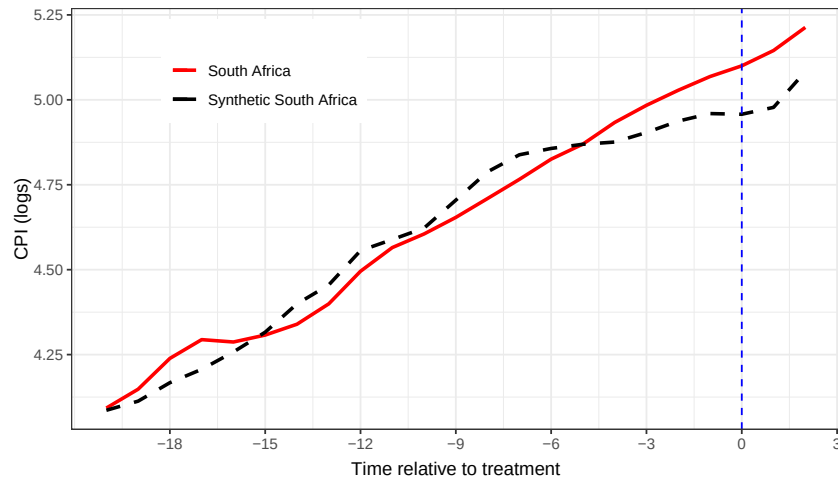
Table A.2: Summary Statistics

	N	Mean	Min	Max
Headline consumer price index	496	135.61	97.00	432.59
Core consumer price index	496	140.41	88.13	395.97
Energy price index	496	130.74	1.20	425.13
food price index	496	151.11	82.22	551.24
Exchange rate	496	1502.01	1.00	14989.86
Real GDP	248	3.56e+08	44735.30	2.93e+09

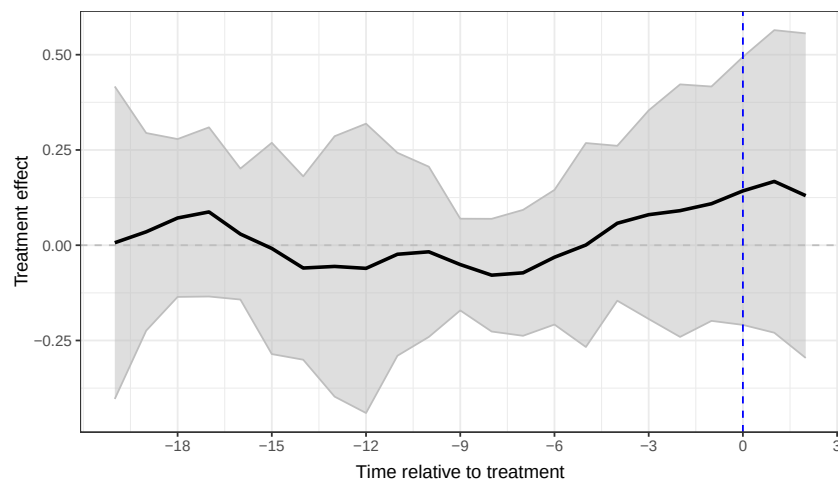
Table A.3: SDiD Estimates of the Average Impact of South Africa's Carbon Tax on Consumer Prices

	Headline	Core	Energy	Food
carbon tax	0.000 (0.023)	-0.005 (0.021)	0.150 (0.121)	-0.043 (0.044)
Observations	248	248	248	248

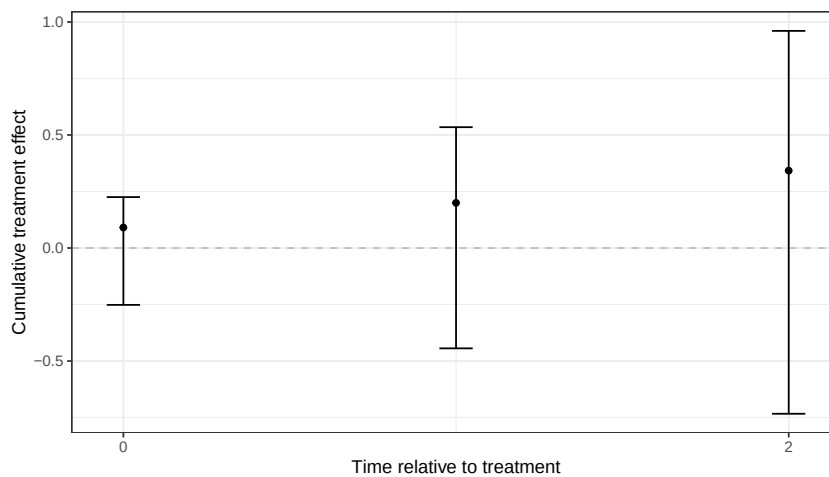
Notes: The table reports the Synthetic Difference-in-Differences estimates of the impact of South Africa's carbon tax on consumer prices. Control variables include real GDP growth and the exchange rate.



(a) Treated and synthetic control



(b) Time Varying Treatment Effect



(c) Cumulative Treatment Effect

Figure A.1: Impact of South Africa's Carbon Tax on Headline Consumer Prices Using Annual Data

Notes: The figure presents the estimated effects of South Africa's carbon tax on the annual headline consumer price index, using the Generalized Synthetic Control Method proposed by Xu (2017). Panel (a) depicts the trajectories of South Africa's headline CPI and its synthetic control; Panel (b) shows the corresponding time-varying treatment effects; and Panel (c) illustrates the cumulative effects. The shaded gray areas and horizontal bars represent the 95% confidence intervals around the estimated effects.

Appendix B: Formulation of the Counterfactual

The main empirical challenge lies in the fact that Y_{jt}^0 —the counterfactual outcome for South Africa after the introduction of the carbon tax—is unobservable. To estimate it, a synthetic control is constructed from a pool of developing countries that did not implement carbon pricing during the study period. Let N_{tr} and N_{co} denote the number of treated and comparison units, respectively. For each country in the donor pool, the outcome equation can be written as:

$$Y_{jt} = \beta X'_{jt} + \lambda_{jt}' f_t + \epsilon_{jt} \quad (.1)$$

Aggregating across all donor countries yields:

$$Y_{co} = \beta X'_{co} + F \Lambda'_{co} + \epsilon_{co} \quad (.2)$$

where Y_{co} and ϵ_{co} are $(T \times N_{co})$ matrices, X_{co} is a $(T \times N_{co} \times k)$ matrix of covariates, and Λ_{co} is an $(N_{co} \times r)$ matrix of factor loadings. The identification of β , F , and Λ_{co} requires that the factors be normalized and orthogonal.

The counterfactual outcome, $\hat{Y}_{jt}^0$, is constructed in three steps:

1. **Estimation of factors and parameters:** Estimate $\hat{\beta}$, $\hat{F}$, and $\hat{\Lambda}_{co}$ using an interactive fixed-effects model (Bai, 2009):

$$\begin{aligned} (\hat{\beta}, \hat{F}, \hat{\Lambda}_{co}) = \underset{\tilde{\beta}, \tilde{F}, \tilde{\Lambda}_{co}}{\operatorname{argmin}} \sum_{j \in C} (Y_j - \tilde{\beta} X_j - \tilde{F} \tilde{\Lambda}_j)' (Y_j - \tilde{\beta} X_j - \tilde{F} \tilde{\Lambda}_j) \\ \text{s.t. } \frac{\tilde{F}' \tilde{F}}{T} = I_r \quad \text{and} \quad \Lambda'_{co} \Lambda_{co} \end{aligned} \quad (.3)$$

2. **Estimation of factor loadings for the treated unit:** For the treated country, estimate $\hat{\lambda}_j$ by minimizing the mean squared prediction error (MSPE) in the

pre-treatment period:

$$\begin{aligned}\hat{\lambda}_j &= \underset{\lambda_j}{\operatorname{argmin}} \sum_{j \in C} (Y_{0j} - \hat{\beta}X_{0j} - \hat{F}_0\lambda_j)' (Y_{0j} - \hat{\beta}X_{0j} - \hat{F}_0\lambda_j) \\ &= (\hat{F}_0'\hat{F}_0)^{-1}\hat{F}_0'(Y_{0j} - \hat{\beta}X_{0j}), \quad j \in C \quad (.4)\end{aligned}$$

3. **Construction of the counterfactual:** Use the estimated parameters to predict the counterfactual outcome for the treated unit:

$$Y_{jt}^0 = \hat{\beta}x_{jt}' + \hat{\lambda}_{jt}'\hat{f}_t \quad (.5)$$

Finally, the estimated average treatment effect on the treated (ATT) is computed as:

$$ATT_t = \frac{1}{N_{tr}} \sum_{j \in C} (Y_{jt}^1 - Y_{jt}^0), \quad t > T_0 \quad (.6)$$